Historical Maps of the Late Medieval & Age of Discovery

An Instructional Cartographic Guide for ArcGIS Story Maps

Five Historical Maps — Volume III

Maps Covered: Carta Marina (Olaus Magnus, 1539) • Fra Mauro's Map of the World (c. 1450) • Abraham Cresques's Map of the World / Catalan Atlas (1375) • The First Map of America (Juan de la Cosa, 1500) • Waldseemüller Map (1507)

Map 1: Carta Marina — 1539 (Olaus Magnus)

The Cartographer

Olaus Magnus (1490–1557)—born Olof Mänsson in Linköping, Sweden—was a Swedish Catholic archbishop, ecclesiastical diplomat, geographer, and historian whose long years of exile in Rome and Venice produced two towering contributions to human knowledge: the Carta Marina of 1539 and the encyclopedic Historia de Gentibus Septentrionalibus (A History of the Northern Peoples, 1555). Dispatched to Rome on a diplomatic mission for the Swedish Crown in 1523, he never returned home, as Sweden's conversion to Lutheranism under King Gustav Vasa made a Catholic bishop's homecoming politically impossible; instead he channeled that exile into the most ambitious and detailed map of northern Europe ever attempted, twelve years in the making.

Historical Facts About the Map

- The full title, Carta Marina et Descriptio Septentrionalium Terrarum ("Marine Map and Description of the Northern Lands"), describes its dual nature precisely. Printed in Venice in 1539 from nine large woodblocks, the assembled map measures approximately 1.7 meters wide by 1.25 meters tall—the largest printed map of any part of Europe at the time. Magnus borrowed 440 ducats to cover printing costs, reflecting both the scale of the production and his personal financial commitment to the project. Only two original copies survive: one at the Bavarian State Library in Munich and one at Uppsala University Library in Sweden.
- The Carta Marina was the first large-scale map to give reasonably accurate coastlines and detailed place names to Scandinavia, Finland, Iceland, Greenland, and the Baltic region, including the Gulf of Bothnia and Gulf of Finland. Magnus drew on personal experience—he had traveled widely across Scandinavia before his ordination—as well as sailor accounts, classical texts, and contemporary voyage reports, resulting in a level of geographic precision for northern Europe that had never before been achieved in print.
- The map’s famous sea monsters—over 40 distinct creatures labeled with letters keyed to a margin legend—were not purely decorative. Magnus presented them as genuine natural history, based on sailor testimony. While many are fantastical or heavily exaggerated, several clearly derive from real animals: the walrus, the orca, the whale, the giant squid, and various shark species are identifiable beneath the medieval embellishments. These illustrations were copied by subsequent European cartographers and appeared on maps of the North Atlantic for more than a century after Magnus drew them.
- Magnus supplemented the Carta Marina with the Historia de Gentibus Septentrionalibus (1555), an 850-page encyclopedia of northern culture, nature, climate, and history in 22 books and 476 chapters. Translated into Italian, German, English, and Dutch within decades of publication, it served as the primary European authority on Scandinavia for over 200 years. Without the Carta Marina and the Historia, northern Europe would have remained a cartographic and cultural blank for the broader European scholarly world throughout the 16th century.

- The map includes one of the earliest cartographic depictions of the whirlpool Maelstrom (labeled Horrenda Caribdis, a reference to the Greek myth of Charybdis), off the coast of northern Norway. Magnus described it as capable of sucking ships to their destruction—a legend that would inspire Edgar Allan Poe's story "A Descent into the Maelström" (1841) and Jules Verne's Twenty Thousand Leagues Under the Sea. The real Moskstraumen, a powerful tidal race off the Lofoten Islands, remains one of the world's strongest tidal currents.

Points of Interest — Scavenger Hunt Style

POI #1 — The Great Sea Serpent

Look for… In the Norwegian Sea west and northwest of the Norwegian coastline, look for a massive serpentine creature labeled with a letter in the margin key—a dragon-like head, enormous coiling body, and a sailing ship nearby dwarfed by its scale.

Magnus described this creature in his Historia as a great sea serpent 200 feet long and 20 feet wide, living in caves along the Norwegian coast and occasionally rising to attack ships. The Stoorworm or Sea Orm was one of the most feared creatures in Norse maritime tradition, and Magnus's detailed depiction—the most authoritative in print at the time—spread the image across European cartography. Subsequent map-makers from Ortelius to Blaeu copied this creature, giving it a 150-year cartographic life.

POI #2 — The Maelstrom (Horrenda Caribdis)

Look for… Off the northwestern coast of Norway, near the islands of the Lofoten archipelago, look for a circular whirlpool symbol labeled in Latin, often accompanied by a small ship being drawn toward the vortex.

Magnus labeled this fearsome whirlpool after the Greek Charybdis—the mythological monster-whirlpool of Homer's Odyssey—linking northern geography to classical mythology in the mind of his educated European reader. The real Moskstraumen tidal race at the Lofoten Islands is one of the world's strongest, and historical accounts of ships being caught in it were genuinely terrifying. This image directly inspired Poe, Verne, and generations of adventure writers.

POI #3 — Mount Hekla in Iceland

Look for… On the large island of Iceland (labeled Islandia) in the upper-left area of the map, look for a mountain with flames or smoke erupting from its peak—the only actively erupting volcano depicted on a European map of this era.

Mount Hekla was believed throughout the Middle Ages and Renaissance to be an entrance to Hell—one of the most widely repeated geographic beliefs in the European scholarly tradition. Magnus depicted it accurately as an active volcano (Hekla has erupted at least 20 times since 874 CE), but the image simultaneously reinforced centuries of supernatural legend. For contemporary European readers, seeing Hekla on the map was seeing the gate of the underworld placed in geographic reality.

POI #4 — Scenes of Reindeer Sledding and Snowshoe Hunting

Look for… On the Scandinavian landmass itself—particularly in the northern regions labeled Lapland and Finnmark—look for small figures of people on snowshoes, hunters in winter gear, and reindeer pulling sleds across snowy terrain.

These are among the earliest detailed European depictions of Sámi and Lappish culture, drawn from Magnus's direct personal observation during his extensive travels across northern Sweden

before 1517. Unlike the generic “exotic peoples” images on many contemporary maps, these figures depict actual practices—reindeer herding, snowshoe travel, and winter hunting—making the Carta Marina one of the most valuable visual ethnographic documents of 16th-century northern life.

POI #5 — The Polar Bear on an Ice Floe

Look for… In the open Arctic waters north of Iceland or Norway, find a white bear depicted standing on a floating ice sheet, isolated in the sea.

Polar bears had occasionally appeared in medieval European royal collections as exotic gifts from Norse Greenland, but their depiction on a geographic map—with accurate geographic placement—was unprecedented. Magnus placed the polar bear at the northern edge of the charted world, signaling that beyond this point lay only ice and the unknown. It was, in effect, a cartographic “Here Be Bears”—a statement about the limits of explored territory.

POI #6 — The Whaling Scene

Look for… In the open North Atlantic or Norwegian Sea, look for an illustration of a large whale being hunted—possibly with small boats, harpoons, and the whale blowing a spout of water.

Whaling was among the most economically important and culturally significant activities in the North Atlantic world, providing oil for lighting, bones for tools, and meat for food across Scandinavia, Iceland, and the Faroe Islands. Magnus's depiction reflected a genuine industry, not mere decoration. The Basques, Norwegians, and Icelanders had developed whaling into a sophisticated practice centuries before it became the global industry of the 17th and 18th centuries.

POI #7 — The Margin Monster Key

Look for… At the lower left or right edge of the map, find a block of text containing numbered or lettered entries corresponding to the sea creatures shown in the ocean areas—a reference key that names and briefly describes each monster.

This key is what transformed the Carta Marina from an illustrated curiosity into a pseudo-scientific natural history document. By assigning each creature a letter and a name, Magnus gave his monsters the scholarly apparatus of a taxonomic catalog. Subsequent cartographers who copied the images also copied the key references, spreading Magnus’s marine zoology across European cartographic tradition as authoritative fact for more than a century.

Web Links

Digital Viewing (Uppsala University Library — all nine sheets, zoomable):
https://www.ub.uu.se/find-and-order/carta-marina/

Print-Optimized Reference (University of Minnesota, James Ford Bell Library — full color):
https://umedia.lib.umn.edu/item/p16022coll184:7550

Map 2: Fra Mauro's Map of the World — c. 1450 (Fra Mauro)

The Cartographer

Fra Mauro (c. 1400–1464) was a Camaldolese monk of the Monastery of St. Michael on the island of Murano, just north of Venice, whose official monastic duties as rent-collector and accountant concealed his true vocation: running one of the most sophisticated cartographic workshops in the medieval world. In his youth, before taking monastic vows, he had traveled widely as a soldier and merchant, gaining first-hand familiarity with the Middle East and eastern Mediterranean that he would draw on throughout his mapping career. A commemorative medal struck in his honor describes him as chosmographus incomparabilis—the incomparable cosmographer.

Historical Facts About the Map

- The Fra Mauro map is universally described as "the greatest memorial of medieval cartography," a circular planisphere drawn on high-quality vellum and set in a gilded wooden frame measuring 2.4 by 2.4 meters (8 by 8 feet). The main circle alone spans over two meters in diameter. It contains more than 3,000 descriptive texts and hundreds of detailed illustrations of cities, mountains, rivers, ships, and trade routes, all rendered in expensive pigments including blue, red, turquoise, green, and gold. It was the most detailed and accurate map of the world that had ever been produced up to that time.
- Two original commissions were made. The first was ordered by King Afonso V of Portugal (completed April 24, 1459, with Fra Mauro's assistant Andrea Bianco—a sailor-cartographer—also receiving payment for his contribution). This copy was sent to Lisbon with a letter from the Doge of Venice to Prince Henry the Navigator encouraging further exploration; it eventually disappeared from São Jorge Castle sometime after 1494. The second copy, made for the Signoria (ruling council) of Venice, was completed by Bianco after Fra Mauro died in 1464 before he could finish it. This is the surviving copy, held today in Venice's Museo Correr.
- The map is oriented with South at the top—following the Arabic cartographic convention of Muhammad al-Idrisi rather than the northern-up Ptolemaic tradition. This inversion was deliberate and philosophically significant: by rejecting the north-up convention that European cartographers had inherited from Ptolemy, Fra Mauro signaled his preference for empirical observation over inherited authority. He wrote directly on the map: "I do not think it derogatory to Ptolemy if I do not follow his Cosmographia," adding that Ptolemy's latitudes had "much terra incognita" that his own sources could correct.
- The map represented a decisive break from medieval theological cartography in several ways: Jerusalem is not placed at the center (Fra Mauro gave a reasoned geographic argument for why it is "somewhat to the west" of center); the Garden of Eden is placed outside the main circle in a corner panel rather than on the earth itself; and the monstrous races that populated the edges of previous mappae mundi are specifically omitted, with Fra Mauro noting that no travelers had ever actually reported seeing them. His decision to privilege eyewitness testimony over classical literary authority was a revolutionary methodological stance.

- The African continent is depicted extending far to the south with an open ocean below it—reflecting early Portuguese reports and Fra Mauro's own reasoned conclusion that the Indian Ocean was not a closed sea and that Africa could be circumnavigated. This map almost certainly influenced the Portuguese court to intensify exploration of the African coast, leading directly to Bartolomeu Dias's rounding of the Cape of Good Hope in 1488. Fra Mauro's African coastline is one of the most consequential geographic predictions in the history of cartography.

Points of Interest — Scavenger Hunt Style

POI #1 — South at the Top — The Inverted Orientation

Look for… Step back and orient yourself: look for Europe in the lower half of the circular map, Africa across the top, and Asia to the right. The map is drawn south-up. Find Italy's boot shape in the lower-center area, pointing upward instead of down.

Fra Mauro's choice to orient his map south-up was a conscious rejection of the dominant Ptolemaic north-up convention in favor of the Arabic tradition he found more geographically useful. It was also a political statement: north-up maps placed northern Europe prominently; south-up maps centered the Mediterranean, Africa, and Asia. For a Venetian cartographer whose city thrived on eastern trade, the world made more sense this way.

POI #2 — The Open Tip of Africa

Look for… On the upper portion of the map (remember, south is at the top), find the continent of Africa—a large landmass. Notice that its southern tip is shown as a rounded, open cape with open sea below and to both sides, rather than the closed land connection to Asia that Ptolemy had drawn.

This is one of the most prophetic details in cartographic history. Every world map based on Ptolemy since the 2nd century CE had shown the Indian Ocean as a closed inland sea surrounded by a southern land bridge connecting Africa to Asia. Fra Mauro, drawing on reports from Portuguese voyagers and Ethiopian monks who had visited Venice, broke from Ptolemy and drew Africa as a navigable peninsula. Twenty-eight years later, Bartolomeu Dias sailed around the Cape of Good Hope, proving Fra Mauro correct.

POI #3 — Japan — One of the First European Depictions

Look for… In the far eastern reaches of the map—the right-hand portion representing Asia—look for an island labeled with something approximating the name of Japan (possibly Cipangu or a variant). This is among the earliest appearances of Japan on a European map.

Fra Mauro drew Japan based on Marco Polo's account of Cipangu—the Island of Gold—lying 1,500 miles off the coast of China. Though his placement was imprecise, its inclusion was significant: it confirmed Japan's existence as a major Asian island in the geographic consciousness of 15th-century Europe, an awareness that would later motivate Columbus (who was partly guided by Fra Mauro's tradition) to sail west in search of exactly this island.

POI #4 — Venice and the Venetian Lagoon

Look for… In the lower-center of the map—the Italian peninsula area—find the city of Venice, depicted with unusual detail compared to other European cities. Look for the characteristic lagoon setting, the bridges, and possibly the bell towers of St. Mark's.

The most accurate area of any cartographer's map is always closest to home, and Fra Mauro's Venice is no exception. As the hub of the Mediterranean trading world and Fra Mauro's own home island, Venice commanded his greatest geographic detail. The wealth of trade knowledge

that flowed through Venetian merchant networks—from Morocco to Persia, from Russia to sub-Saharan Africa—was the raw material from which the entire map was built.

POI #5 — The Four Corner Panels Outside the Main Circle

Look for… At the four corners surrounding the circular main map, look for four smaller diagrams: a Ptolemaic solar system, the four elements (earth, water, fire, air), the Garden of Eden, and a diagram of the Earth as a globe.

These four panels reveal Fra Mauro's intellectual breadth and his commitment to presenting geography within a complete cosmological framework. By placing the Ptolemaic cosmos, the four Aristotelian elements, and the Globe alongside his empirical world map, he was arguing that observation and classical philosophy were complementary, not contradictory. Crucially, placing Eden in a separate corner panel rather than on the earth itself was a quiet but revolutionary statement: Paradise is not a geographic location. It is a theological concept.

POI #6 — The 3,000+ Text Annotations

Look for… Look across any portion of the map for the dense clusters of handwritten text—tiny inscriptions in Italian and Latin filling every region of the circular image. Some are city names; many are paragraph-length descriptions of customs, trade goods, rivers, and legends.

The text annotations are what transform the Fra Mauro map from a geographic document into an encyclopedia. Each one represents a specific piece of knowledge acquired from a named source—a Venetian merchant, a returning pilgrim, an Ethiopian monk, a Portuguese navigator—and Fra Mauro often identifies his source explicitly as validation. This citation practice made the map one of the earliest examples of evidentiary scholarship in European cartography: not just where things are, but how we know where they are.

POI #7 — The Indian Ocean Trade Route Network

Look for… In the central-right portion of the map, find the Indian Ocean—a large body of water shown open at its southern end, populated with labeled islands, coastal cities, and possibly small ship illustrations. Look for the Red Sea and Persian Gulf opening into it.

The detail of the Indian Ocean on Fra Mauro's map reflects Venice's vital commercial interest in the eastern trade routes. Indian Ocean trade—spices from the Moluccas, silk from China, cotton from India, gold from East Africa—flowed through Arab middlemen to Venetian merchants, and controlling or understanding these routes was worth fortunes. Fra Mauro's Indian Ocean is the most commercially important body of water on the map, and it shows.

Web Links

Digital Viewing (Museo Correr, Venice — interactive high-resolution exploration platform): https://fra-mauro.it/en/

Print-Optimized Reference (Ruderman Rare Maps, full-size facsimile description): https://raremaps.com/gallery/detail/89395/facsimile-of-the-circa-1450-fra-mauro-mappamundi-mauro

Map 3: Abraham Cresques' Map of the World (Catalan Atlas) — 1375

The Cartographer

Abraham Cresques (1325–1387) was a Jewish cartographer, instrument-maker, and manuscript illuminator from Palma, Majorca, a cosmopolitan island where Arabic, Hebrew, and Catalan geographic traditions intersected in a unique scholarly environment. He served as official mapmaker to King Pedro IV of Aragon—who called him "master of world maps and compasses"—and worked alongside his son Jehuda Cresques in a workshop that represented the absolute pinnacle of 14th-century cartographic achievement. Working fluently across three cultural and linguistic traditions (Christian Catalan, Jewish Hebrew, and Islamic Arabic), Cresques was uniquely positioned to synthesize the geographic knowledge of three civilizations into a single coherent vision of the world.

Historical Facts About the Map

- The Catalan Atlas was created in 1375 as a diplomatic gift from Prince John of Aragon (the future King John I) to his cousin Charles VI of France. Cresques and his son were paid 150 Aragonese gold florins and 60 Mallorcan pounds for the commission—a significant sum reflecting the atlas's status as a luxury object of state. The atlas consists of six vellum leaves richly painted with pigments, gold, and silver: the first two contain cosmographic, astronomical, and astrological texts, while the remaining four form the map itself, stretching nearly 10 feet wide when unfolded. It is now held by the Bibliothèque nationale de France in Paris (Ms. Espagnol 30).
- The Catalan Atlas was the first cartographic work to systematically combine the navigational precision of portolan charts with the encyclopedic breadth of medieval mappae mundi, effectively creating the template for the illustrated world atlas. It was the first map to feature an ornate compass rose (rather than a simple wind diagram), and it incorporated knowledge drawn from Marco Polo's Travels, the voyage accounts of Sir John Mandeville, Arab and Jewish geographic texts, and the most current portolan chart data available in Majorca—the most cosmopolitan cartographic center in the Mediterranean world.
- The atlas's depiction of Africa, Asia, and the trade world was unprecedented in its detail and ambition. West Africa was shown with the routes of the trans-Saharan gold trade; Central Asia with the cities of the Silk Road; China and the Far East with details drawn directly from Marco Polo; and the Indian Ocean with a network of named ports reflecting 14th-century commercial geography. For the first time on a European map, the known world was presented as an interconnected commercial system rather than a series of separate regional spaces.
- The aftermath of the Catalan Atlas is one of cartographic history's most poignant stories. In 1391, anti-Jewish riots swept across the Crown of Aragon. Abraham Cresques had died in 1387, but his son Jehuda was forcibly baptized and renamed Jaume Ribes, and eventually relocated to Portugal where he may have served as cartographic adviser to the early Portuguese exploration program under Prince Henry the Navigator—a direct

line from the greatest medieval world map to the systematic exploration of the African coast and beyond.

- The Catalan Atlas established the tradition of the beautifully illustrated, commercially oriented world atlas that would dominate European cartography for the next two centuries—through Dulcert, Bianco, Benincasa, and ultimately through the great printed atlases of Ortelius and Mercator. Its combination of scientific accuracy, artistic richness, and commercial intelligence defined what a world map was supposed to be.

Points of Interest — Scavenger Hunt Style

POI #1 — Mansa Musa — Emperor of Mali on His Golden Throne

Look for… In the interior of West Africa (the lower-left area of the world map panels), find a richly illustrated seated figure wearing a golden crown and holding a gold orb or nugget, extending it toward a turbaned camel rider approaching from the desert. He sits on an elaborate throne.

This is Mansa Musa (r. 1312–1337), whose 1324 hajj pilgrimage to Mecca became the most famous display of wealth in medieval history—he reportedly distributed so much gold in Cairo that he triggered severe inflation across Egypt for over a decade. Cresques's inscription (referring to him as "Musse Melly, lord of the Negroes of Guinea") was the most detailed European cartographic acknowledgment of West African power and wealth up to that point, and it planted the seed of Portuguese interest in sailing south along the African coast to find the source of all that gold.

POI #2 — The First Ornate Compass Rose

Look for… In the open sea areas of the map—probably the western Mediterranean—find a large, decorative starburst or circular design with many radiating points, rendered in multiple colors with artistic embellishment. This is the first ornate compass rose in cartographic history.

Earlier portolan charts used simple wind diagrams or plain rhumb-line networks. Cresques transformed the navigational compass into a work of art, giving it the visual richness of an illuminated manuscript ornament while preserving its functional geometry. This innovation merged science and beauty in a way that defined map aesthetics for the next three centuries—every decorative compass rose on every subsequent map descends from this one.

POI #3 — Marco Polo's Asia — Cathay and the Silk Road Cities

Look for… In the eastern panels of the atlas, find a detailed depiction of Central Asia, India, and China (labeled Cathay), populated with small illustrations of rulers on thrones, elephants, camels, and text references to specific cities and kingdoms drawn from Marco Polo's accounts.

Cresques drew more extensively from Marco Polo's Travels than any previous cartographer, making the Catalan Atlas the most geographically comprehensive depiction of Asia in medieval European cartography. The Silk Road cities—Samarkand, Kashgar, Khanbaliq (Beijing)—appear with their commercial character identified. For European merchants who had lost direct access to these routes after the Mongol Empire collapsed, this map was both a memory of what had been possible and a vision of what might be regained.

POI #4 — The Camel Caravans of the Sahara

Look for… Across the Saharan interior of northern Africa, look for small illustrations of camel caravans—strings of camels carrying loads, guided by turbaned riders—crossing featureless desert terrain.

These caravans represent the trans-Saharan gold and salt trade that underpinned the wealth of the Mali and Songhai empires and connected sub-Saharan Africa to the Mediterranean world. This was the most important commercial route in medieval Africa, and Cresques depicted it with the same commercial intelligence that defined the portolan tradition: knowing where the trade routes ran was as important as knowing where the coastlines ran. The caravans literally map the flow of gold from the mines of West Africa to the markets of North Africa and Europe.

POI #5 — The Astronomical and Cosmological Opening Panels

Look for… On the first two leaves of the atlas (the leftmost panels when fully laid out), look for circular diagrams showing the phases of the moon, the zodiac, and planetary arrangements—accompanied by dense text in Catalan explaining the cosmic order.

Cresques opened his atlas not with geography but with astronomy, because in the 14th-century understanding of the world, you could not comprehend where you were on earth without understanding your position in the cosmos. The lunar calendar panels were also practically useful: sailors used the phases of the moon to plan tides and passages. By combining cosmology, calendar, and geography in a single bound object, Cresques created the first unified reference system for navigation—the intellectual ancestor of every almanac and nautical table that followed.

POI #6 — The Iberian Peninsula and Mediterranean Coastlines

Look for… On the western panels of the atlas, trace the detailed coastlines of Spain, Portugal, Italy, and the western Mediterranean, dense with labeled ports shown as small colored rectangles or tower symbols, with national flags identifying their political allegiance.

This section of the atlas represents the portolan tradition at its most precise: these coastlines were not derived from literary sources but from generations of Mediterranean sailor observations, refined and corrected through the competitive Majorcan cartographic school. The density of port labels and the accuracy of coastline angles reflect a working navigational document as much as a decorative map—proof that Cresques could unite the art of illustration with the science of practical seamanship.

POI #7 — The Three Magi and Eastern Christianity

Look for… In the Asian panels, look for a group of crowned or robed figures on horseback identified as the biblical Magi—the Three Kings—traveling westward from their homeland in Persia or Central Asia.

The inclusion of the Magi's homeland reflects the medieval European belief that Eastern Christianity was a living political reality. The legend of Prester John—a mythical Christian king ruling in Asia or Africa—was taken seriously by popes and kings who hoped to find Christian allies in the East for a new Crusade. For Cresques, showing the Magi's origin on the map was a theological as much as geographic statement: the whole world, even its most distant corners, had been touched by Christ's story.

Web Links

Digital Viewing (Bibliothèque nationale de France, Gallica — full high-resolution all panels): https://gallica.bnf.fr/ark:/12148/btv1b55002481n

Print-Optimized Reference (Smarthistory scholarly essay with annotated panel images): https://smarthistory.org/catalan-atlas/

Map 4: The First Map of America — 1500 (Juan de la Cosa)

The Cartographer

Juan de la Cosa (c. 1450–1510) was a Castilian-Basque navigator, cartographer, and explorer from Santoña in Cantabria who has the singular distinction of being both the owner-captain of Columbus's flagship Santa María on the 1492 voyage and the maker of the oldest surviving European map to depict the Americas. Over seventeen years he undertook seven Atlantic voyages, sailing with Columbus, Ojeda, Vespucci, and Bastidas, exploring the Caribbean, Central America, and the northern coast of South America. He died on his seventh and final voyage in 1510, killed by a poisoned arrow near present-day Turbaco, Colombia, while attempting to establish a Spanish settlement—a violent end that was the direct consequence of the colonial enterprise his own map had helped to chart.

Historical Facts About the Map

- The map bears a self-identifying inscription in Spanish: "Juan de la Cosa made this in the port of Santa María in the year 1500." Puerto de Santa María was a small port near Cádiz, Andalusia. The map is drawn on two joined sheets of parchment sewn onto a canvas backing, measuring 96 cm high by 183 cm wide—about the size of a large desk. It is a portolan chart in style, with rhumb lines, a compass rose, and coastal notation, but it extends far beyond the Mediterranean to encompass a full world map including the first known cartographic rendering of the American coastlines.
- The map is the only known cartographic work made by a direct eyewitness of Columbus's first voyages, and it incorporates knowledge from multiple competing Atlantic explorations: Columbus's four voyages, John Cabot's 1497 voyage to Newfoundland (shown with the English flag), Alonso de Ojeda's 1499 South American coastal survey (on which de la Cosa served as pilot), Vicente Piñzón's voyage, and Pedro Álvares Cabral's 1500 landfall in Brazil. It is in this sense a comprehensive intelligence summary of everything Europe knew about the Atlantic world as of the year 1500.
- After being classified as a state secret by the Spanish Crown (the geographic intelligence it contained was of enormous strategic value), the map disappeared from public knowledge for more than 300 years. It resurfaced in a Paris junk shop in 1832, where it was purchased by the French scientist and cartographic historian Charles Athanase Walckenaer for a trivial sum. Alexander von Humboldt, who examined it shortly after its rediscovery, was the first modern scholar to identify its true significance. The Spanish government purchased it in 1853; it is now in the Museo Naval in Madrid.
- The map famously depicts Cuba correctly as an island—contradicting Columbus's dying insistence that Cuba was a peninsula of the Asian mainland. De la Cosa had sailed along Cuba's coast himself and knew it was an island; by drawing it that way, he took a silent but definitive stand against the Admiral's geographic conviction. This detail makes the map a document of intellectual courage as much as geographic knowledge.
- The region of Central America is covered with a large image of Saint Christopher carrying the Christ child across the water—occupying exactly the area where a passage

to the Indian Ocean would need to exist if Columbus's theory of reaching Asia by sailing west were correct. Scholars have interpreted this image both as devotional imagery (Christopher = Christ-bearer, honoring Columbus's Christian mission) and as a deliberate cartographic evasion: by placing a saint over the unknown western regions, de la Cosa left open the possibility of an Asian passage without having to commit to a geographic claim he could not verify.

Points of Interest — Scavenger Hunt Style

POI #1 — The Americas as a Green Landmass

Look for… On the left half of the map, find the large area rendered in vivid green—unlike any other region on the map. This is the American continent, shown in the lush color reportedly used to evoke the tropical vegetation encountered by European voyagers.

The green color of the Americas on de la Cosa's map is historically significant: it was a deliberate visual choice to communicate the extraordinary fertility and vegetation that Spanish explorers described upon first encountering Caribbean landscapes. The Americas were unlike anything in the European or Mediterranean geographic imagination—a world of overwhelming green abundance. Making them literally green on the map communicated this radical otherness to anyone who picked up the chart.

POI #2 — Cuba Drawn as an Island

Look for… In the Caribbean area of the map, find the island of Cuba—rendered with some accuracy as a recognizable island shape. Notice that it is completely surrounded by water, separated from the mainland.

This detail was politically explosive. Christopher Columbus had insisted until his death (which came six years after this map was made) that Cuba was a peninsula of the Asian mainland, and he had even forced his crew to sign sworn affidavits to that effect on his second voyage. De la Cosa, who had sailed along Cuba's coast himself, knew this was false—and drew it truthfully. The map was a quiet but definitive correction of the Admiral's most cherished geographic conviction.

POI #3 — Saint Christopher Covering Central America

Look for… In the western portion of the Americas, where the map transitions from the Caribbean coast to unknown territory, find a large illustration of Saint Christopher—a robed giant carrying the Christ child across a body of water, dominating the area where Central America and any possible passage to Asia would be located.

This image is one of the most discussed decisions in the history of cartography. By placing a saint over the unknown western regions, de la Cosa simultaneously honored Columbus's Christian mission, avoided having to make a geographic claim he could not verify, and left open the possibility that a western passage to Asia might still exist. It was cartographic diplomacy—knowledge management as political art.

POI #4 — Flags of Competing European Crowns

Look for… Along the coastal areas of the Americas, look for small national flags or standards—the Castle and Lion of Castile and León for Spanish claims, and the English flag of St. George along the northern coastline of what is now Canada and New England.

The flags on this map made it a political document as well as a geographic one. The English flag in the north reflects John Cabot's 1497 voyage to Newfoundland under Henry VII—knowledge de la Cosa incorporated from intelligence sources in the competing courts of

Europe. By showing both Spanish and English claims, the map documented the fact that the race for Atlantic empire was already multi-national in 1500, with competing crowns staking claims to regions that none of them had yet seen in full.

POI #5 — The Compass Rose and Rhumb Line Network

Look for… At a prominent location in the map—probably the Atlantic Ocean area—find the large compass rose and the radiating network of rhumb lines crossing the entire chart in multiple directions.

Despite its extraordinary subject matter, de la Cosa's map is structured as a working portolan chart—the navigational tradition he knew best from his years as a Mediterranean and Atlantic pilot. The rhumb line network meant that the chart could, in principle, be used for actual navigation: a pilot could find his bearing from any point on the map to any other. This practical orientation distinguished the de la Cosa map from decorative or theoretical world maps and signaled its intended audience: professional sailors, not scholars.

POI #6 — The African Cape and Indian Ocean

Look for… On the right side of the map, find the triangular southern tip of Africa—the Cape of Good Hope—with the Indian Ocean opening to the east. Look for Portuguese place names along the African east coast and extending toward India.

Vasco da Gama had rounded the Cape of Good Hope in 1498, just two years before this map was made, and de la Cosa incorporated the new Portuguese geographic intelligence almost immediately. The Indian Ocean section of the map reflects the first fruits of the Age of Exploration's eastern arm, showing that by 1500, European geographic knowledge had expanded in two directions simultaneously: west toward the Americas and east around Africa toward India.

Web Links

Digital Viewing (Google Arts & Culture, Museo Naval Madrid — high resolution story): https://artsandculture.google.com/story/chart-of-juan-de-la-cosa-the-first-known-map-of-america-fundaci%C3%B3n-museo-naval/CgVBiowwoJqZIQ

Print-Optimized Reference (Factum Foundation — 3D-scanned facsimile viewer): https://factumfoundation.org/our-projects/digitisation/the-spanish-gallery-at-bishop-auckland/the-map-of-juan-de-la-cosa/

Map 5: The Waldseemüller Map — 1507 (Martin Waldseemüller)

The Cartographer

Martin Waldseemüller (c. 1470–1521) was a German cartographer, humanist scholar, and cosmographer who worked at the Gymnasium Vosagense (the Vosgean Gymnasium) in Saint-Dié-des-Vosges, in the duchy of Lorraine—then part of the Holy Roman Empire. Operating in a collaborative scholarly environment funded by Duke René II of Lorraine, he worked alongside the humanist poet Matthias Ringmann (who may have been the intellectual author of the decision to name the new continent “America”) and a circle of translators and engravers to produce one of the most consequential cartographic documents in human history. His 1507 world map gave a name to two continents that would be home to nearly a billion people within four centuries.

Historical Facts About the Map

- The full title of the Waldseemüller map is Universalis Cosmographia Secundum Ptholomaei Traditionem et Americi Vespucii Alioru[m]que Lustrationes ("A Drawing of the Whole Earth Following the Tradition of Ptolemy and the Voyages of Americus Vespuccius and Others"). Printed in April 1507 from twelve woodblock sheets in a reported edition of 1,000 copies, the assembled map measures approximately 4.5 by 8 feet—the largest printed world map of the 16th century up to that date. Of the original 1,000 copies, only one complete example is known to survive today. It was discovered in 1901 in the library of Prince Waldburg-Wolfegg in Schloss Wolfegg in Württemberg, Germany, and purchased by the Library of Congress in 2003 for $10 million—the highest price ever paid for a single map.
- The map is the first in history to use the name “America,” placed on what we now call South America in the map’s third-from-left horizontal panel. The name honored Amerigo Vespucci, the Florentine explorer whose letters—published in Europe from 1503 onward—argued convincingly that the lands Columbus had reached were not the coast of Asia but a “New World” (Mundus Novus) entirely unknown to the ancients. In the companion publication Cosmographiae Introductio, Waldseemüller explained: “Now, truly, since these parts have been more extensively explored, and another fourth part has been discovered by Americus Vespuccius... I do not see why anyone would rightly forbid naming it Amerigen, or America, that is, the land of Americus.”
- The map made two additional world-historical firsts: it was the first map to depict the Americas as a separate hemisphere entirely surrounded by ocean (rather than as an extension of Asia), and it was the first to show a distinct ocean to the west of the Americas—what we now call the Pacific. Neither Vasco Núñez de Balboa (who first saw the Pacific from the Isthmus of Panama in 1513) nor any other European had actually crossed to the Pacific coast of the Americas by 1507. Waldseemüller appears to have inferred the Pacific’s existence from Vespucci’s accounts and from the internal logic of the geography, making this map one of the most audacious acts of geographic inference in history.

- Curiously, Waldseemüller apparently had second thoughts. His 1513 Ptolemy edition maps dropped the name “America” entirely, labeling the same region “Terra Incognita.” His 1516 large wall map (also called Carta Marina, not to be confused with Olaus Magnus’s) labeled North America “Terra de Cuba Asie Partis”—Land of Cuba, Part of Asia—reverting to the Columbian conception he had rejected in 1507. But by then it was too late: the name “America” had been adopted by other cartographers, confirmed by Gerardus Mercator in 1538, and had become permanent.
- The map also contained printed portraits of Ptolemy (representing the Old World geographic tradition) and Amerigo Vespucci (representing the New) at the top corners—a visual manifesto declaring that a new age of geography had begun, one that would be built on Ptolemy’s mathematical framework but enriched by the empirical discoveries of the Age of Exploration. The Library of Congress has called it “America’s birth certificate,” a description that captures both its legal significance (the first legal use of the name) and its symbolic weight.

Points of Interest — Scavenger Hunt Style

POI #1 — The Word “AMERICA” on the Southern Continent

Look for… In the left-center portion of the map—on the long, narrow landmass that represents South America—look for the single Latin word “AMERICA” written across the interior of the continent in large capital letters.

This is the most consequential place name in the history of cartography. A continent that had existed for hundreds of millions of years and been inhabited for at least 15,000 years received its European name—in print, for the first time—at this exact location. The name was chosen by a German humanist scholar in a small French mountain town who had never seen the continent and never would, based on letters attributed to a Florentine explorer. That this arbitrary naming became permanent is one of history’s most remarkable accidents.

POI #2 — The Pacific Ocean— Inferred Before It Was Seen

Look for… Between the western coast of the Americas and the eastern coast of Asia, look for an open body of water—a distinct ocean separating the two continents. Note that this ocean is wider than the Atlantic on the map.

The Pacific Ocean—the largest geographic feature on earth—was not seen by any European until Vasco Núñez de Balboa crossed the Isthmus of Panama in 1513, six years after this map was printed. Yet here it is, on the Waldseemüller map of 1507, correctly depicted as a distinct ocean wider than the Atlantic. Whether Waldseemüller inferred it from Vespucci’s calculations, from undocumented Portuguese intelligence, or from pure geographic logic remains one of the great unsolved mysteries of cartographic history.

POI #3 — Portraits of Ptolemy and Vespucci at the Top

Look for… At the very top of the map, in the decorative border panels flanking the polar projection, find two portrait busts: on the left, an older bearded man holding a T-O diagram—representing Ptolemy—and on the right, a contemporary Renaissance figure holding a navigational instrument—representing Amerigo Vespucci.

This deliberate pairing was Waldseemüller’s visual argument for how the new geography worked: Ptolemy provided the mathematical and philosophical framework; Vespucci provided the empirical observations that extended and corrected it. In 1507, Ptolemy was still the foundational authority for all geographic thinking, and naming him alongside Vespucci was a

way of legitimizing the new discoveries within the classical tradition rather than rejecting it outright.

POI #4 — North America as Terra Incognita

Look for… In the left portion of the map, above the labeled “America” of South America, find a smaller northern landmass with little interior detail, labeled “Terra Incognita” (Unknown Land) or sparse text. Notice the gap or strait separating it from South America.

Waldseemüller had very little reliable information about North America in 1507—far less than he had about South America, whose eastern coast Vespucci had extensively surveyed. The depiction of North America is sparse, speculative, and almost blank by comparison—a cartographic admission of ignorance that stands in striking contrast to the confident naming of South America directly below it. This contrast captures the uneven pace of geographic discovery: south was known, north was not yet.

POI #5 — The Twelve-Sheet Structure of the Wall Map

Look for… Look for the faint lines dividing the map into twelve rectangular sections—the grid of the woodblock printing. The map was printed from twelve separate blocks, each approximately 18 x 24.5 inches, and assembled by the purchaser.

The twelve-sheet format was both a technical achievement and a commercial strategy. Printed maps at this scale required enormous woodblocks, extraordinary precision in registration, and a distribution system capable of shipping twelve large sheets flat and undamaged. The fact that Waldseemüller’s workshop in a small French mountain town achieved this in 1507 reflects the extraordinary ambition and organizational sophistication of Renaissance humanist scholarship. Only one complete copy survived because most purchasers assembled and displayed their maps on walls where they eventually deteriorated.

POI #6 — The Strait Separating North and South America

Look for… Between the northern and southern American landmasses, look for a narrow water passage—an inferred strait separating what would later be called North and South America.

Waldseemüller depicted a strait separating the two American landmasses—a geographic feature that did not appear on the Juan de la Cosa map and that did not actually exist in the form shown. This phantom strait reflects Waldseemüller’s attempt to reconcile different voyage reports and theoretical calculations into a single coherent geography. Ironically, by inventing a separation between the two continents, he was closer to the geographic reality than Columbus ever was: North and South America are indeed separate landmasses, joined only by the narrow Isthmus of Panama.

POI #7 — The Old World in the Right Panels— Europe, Africa, and Asia

Look for… On the right half of the map, find the familiar outlines of Europe, Africa, and Asia—depicted with varying accuracy. Notice that Africa’s southern tip is shown as navigable, and that Asia extends far to the east with a recognizable (if imperfect) peninsula of India.

Waldseemüller’s Old World was built on the best available Portuguese and Ptolemaic sources of 1507, and it shows: the Mediterranean is accurate, Africa is well-rendered (reflecting Portuguese survey data from da Gama’s voyages), and India is recognizable. But Asia becomes increasingly speculative toward the east, and the relationship between Asia and the Americas on the far right and far left of the map was deliberately left ambiguous—the honest cartographic acknowledgment that the full circuit of the globe was not yet understood.

Web Links

Digital Viewing (Library of Congress — official high-resolution interactive viewer, the surviving original): https://www.loc.gov/item/2003626426/

Print-Optimized Reference (Library of Congress Information Bulletin article with full scholarly context): https://www.loc.gov/loc/lcib/0309/maps.html

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Notes on Using These Maps in ArcGIS Story Maps

Each Point of Interest entry in this guide is formatted for direct use as a callout panel in an ArcGIS Story Map. Copy the POI title as your panel header, use the “Look for…” sentence as your visual prompt for viewers, and follow it with the historical significance paragraph as your body text. For the best interactive experience, pair each POI callout with a cropped or zoomed screenshot of the relevant map area.

All five maps in this volume are available in high-resolution digital form through the linked institutional repositories, most of which permit educational and non-commercial use explicitly. The Library of Congress Waldseemüller viewer and the Fra Mauro interactive platform at fra-mauro.it are particularly well suited to the zoomed detail work that ArcGIS Story Maps panels require.

NOTE: The Carta Marina (Olaus Magnus, 1539) appears in both this volume and Volume II of the series. The coverage here is self-contained and can be used independently. Readers wishing to cross-reference the two entries will find complementary details in each.